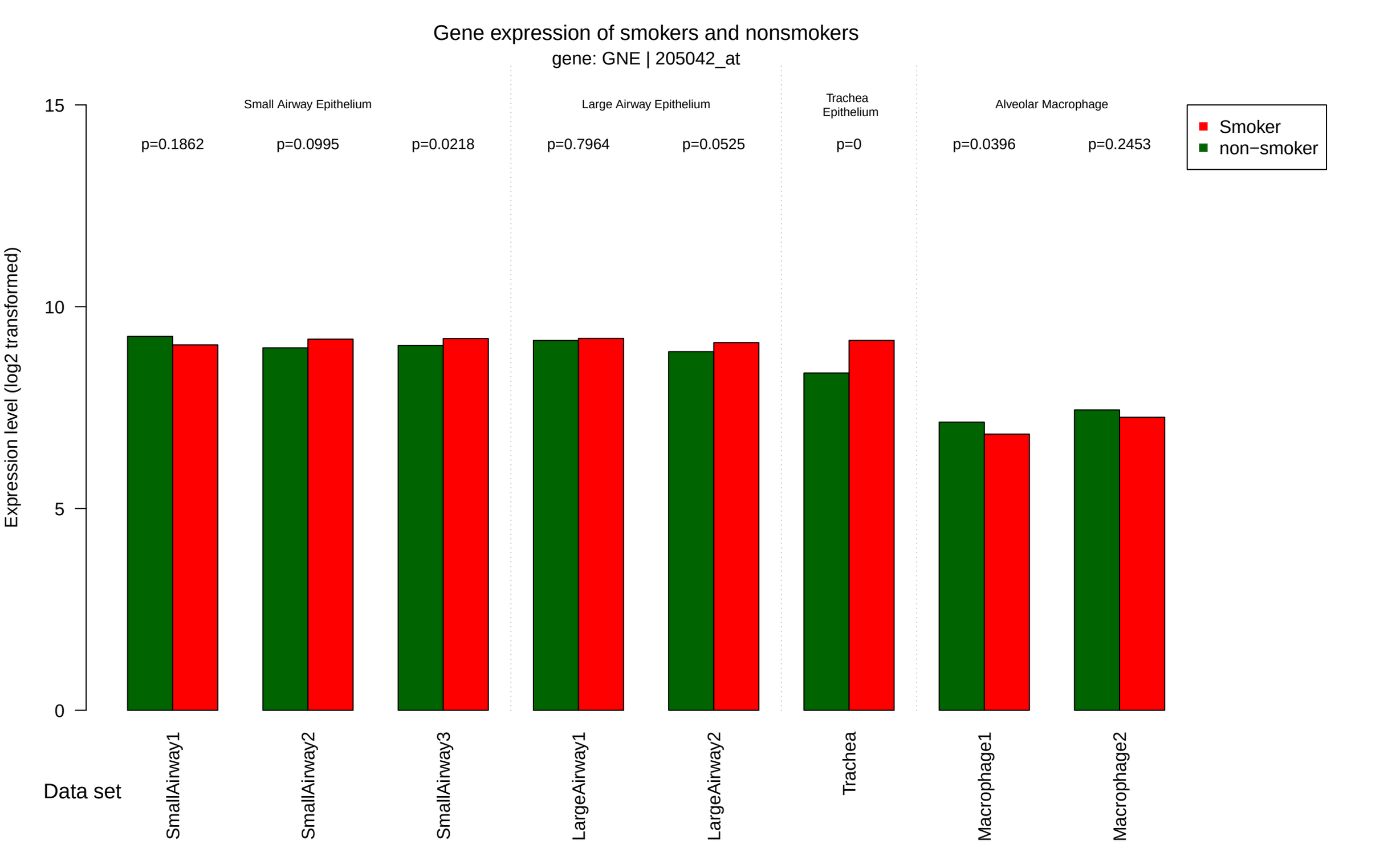
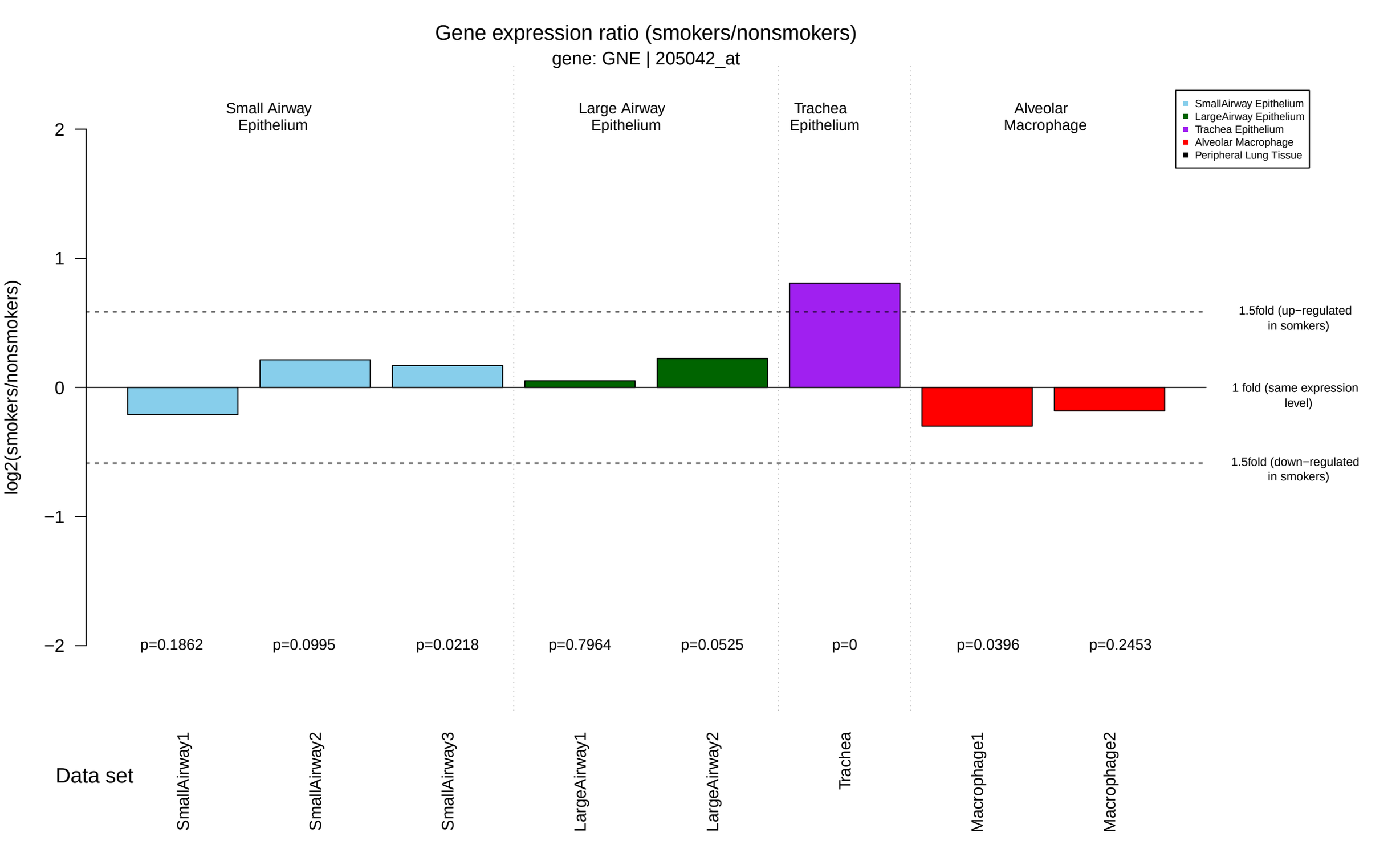
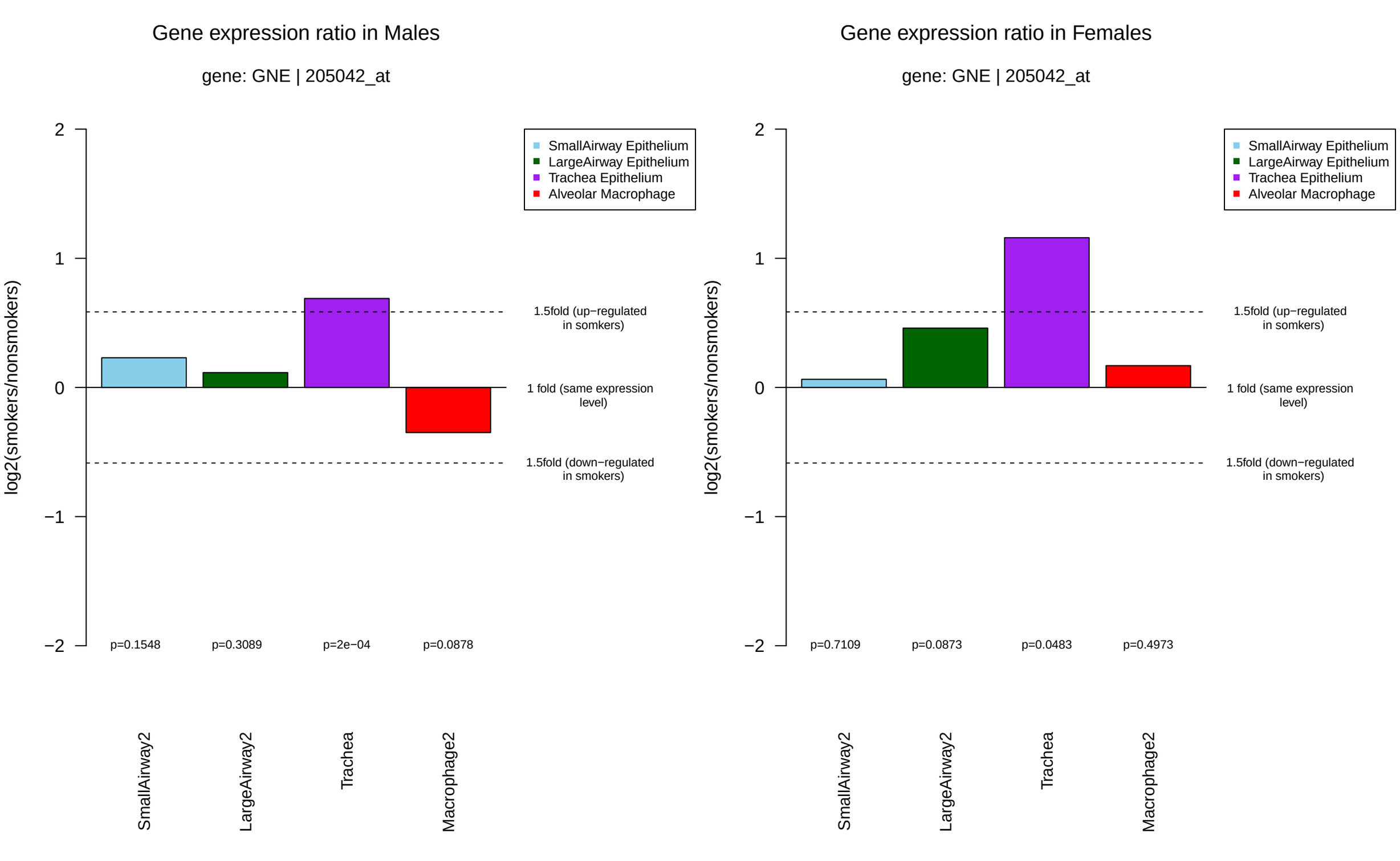
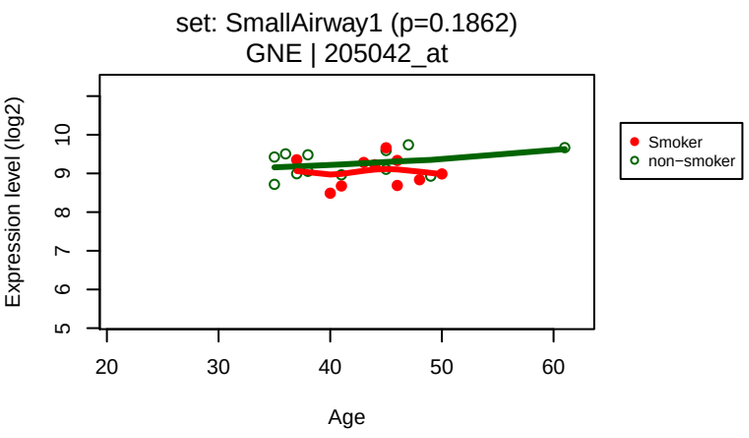
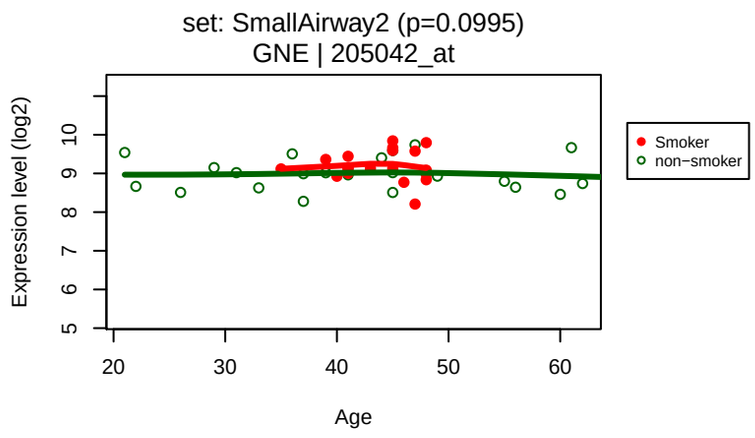
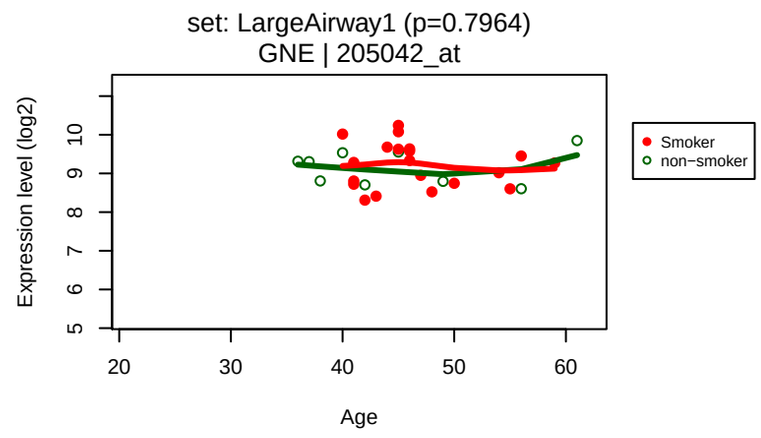

Gene expression of smokers and nonsmokers
gene: GNE | 205042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1862
p=0.0995
p=0.0218
p=0.7964
p=0.0525
p=0
p=0.0396
p=0.2453
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

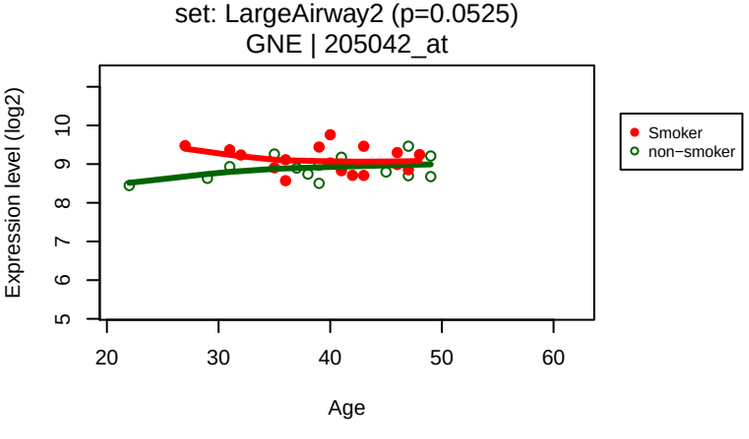

Gene expression ratio (smokers/nonsmokers)
gene: GNE | 205042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1862
p=0.0995
p=0.0218
p=0.7964
p=0.0525
p=0
p=0.0396
p=0.2453
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

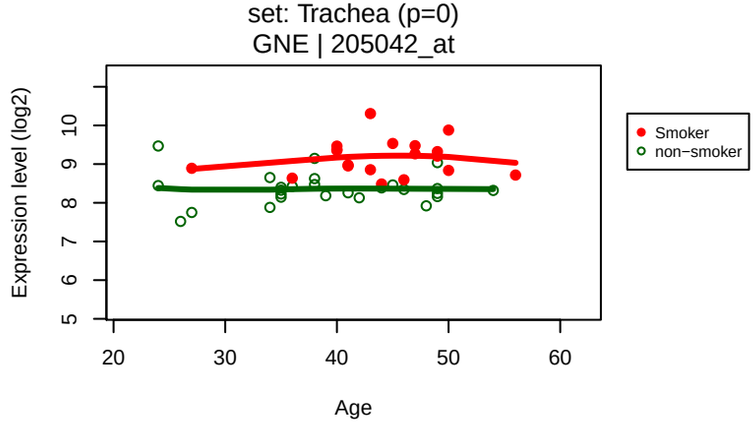

Gene expression ratio in Males
gene: GNE | 205042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1548
p=0.3089
p=2e−04
p=0.0878
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNE | 205042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7109
p=0.0873
p=0.0483
p=0.4973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

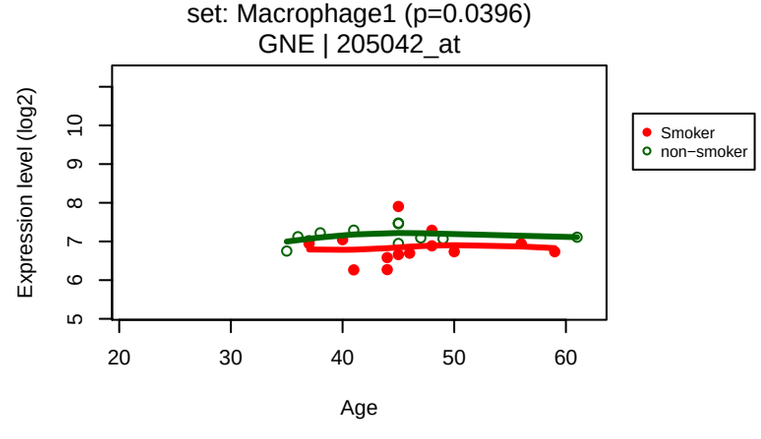

set: SmallAirway1 (p=0.1862)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker

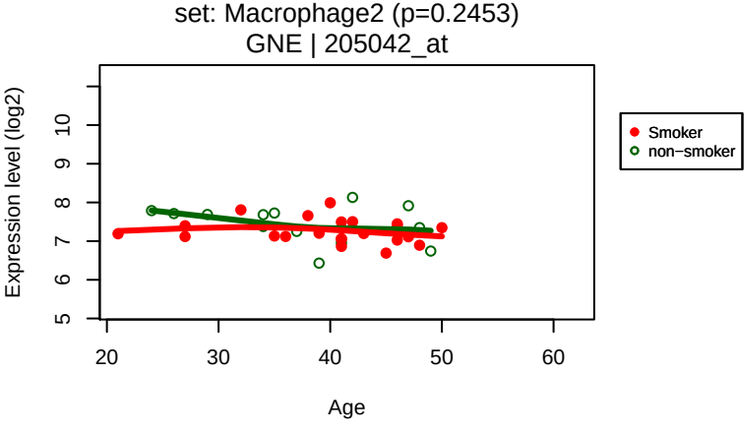

set: SmallAirway2 (p=0.0995)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker

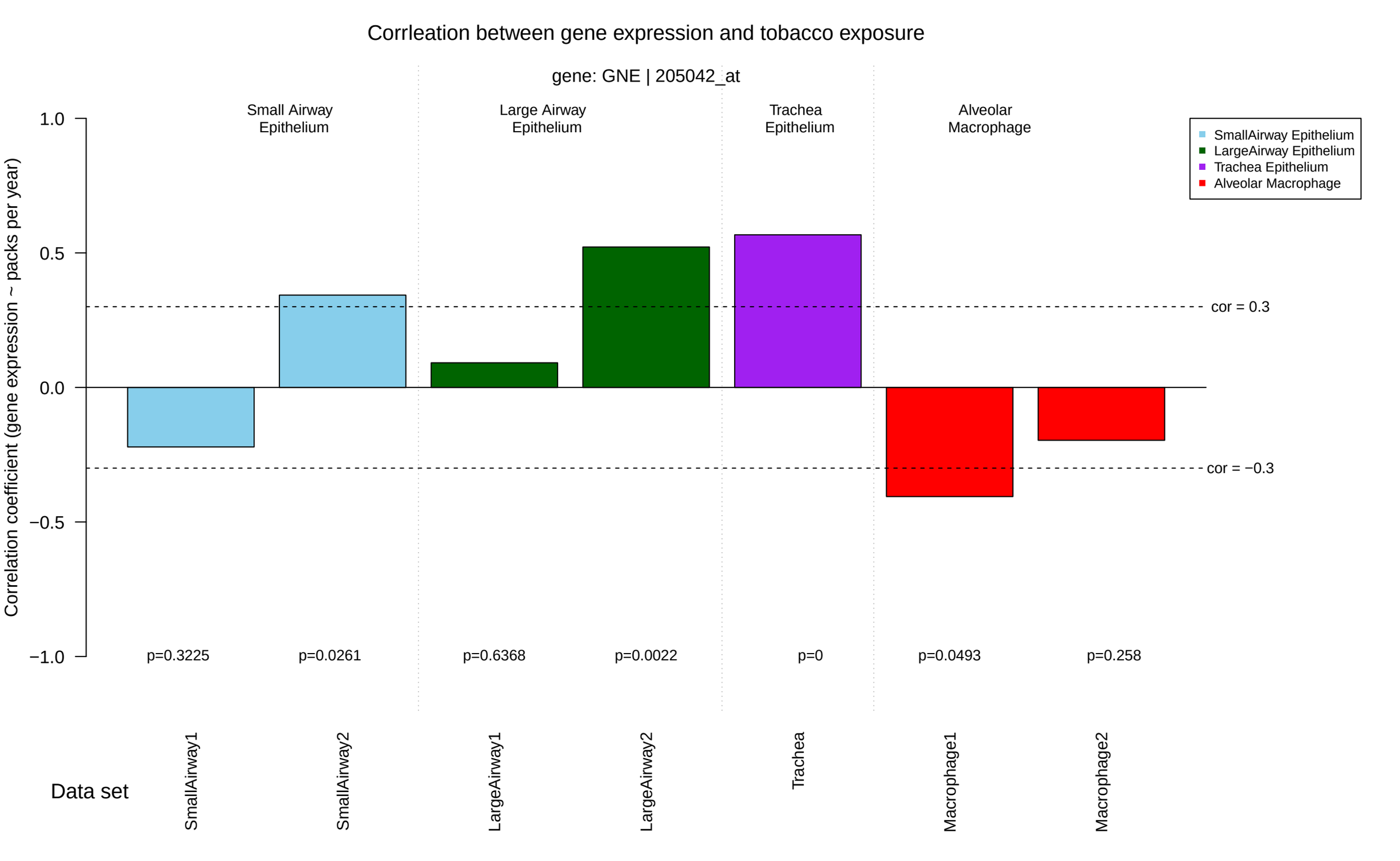

set: LargeAirway1 (p=0.7964)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0525)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0396)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2453)
GNE | 205042_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNE | 205042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3225
p=0.0261
p=0.6368
p=0.0022
p=0
p=0.0493
p=0.258
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2